

5-Mark Questions & Answers

Q1. Explain the three-address, two-address, one-address, and zero-address instruction formats with examples and a comparison table.

1. Instruction Format Types

1. Three-Address Format: Specifies Opcode, Destination, and two Source Operands.

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2. Comparison Table

| Format | Explicit Operands | Implicit Location | Instruction Length | Program Length (No. of Insts) |

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Q2. Explain the basic instruction cycle with interrupt processing, including the interrupt processing cycle steps.

1. Combined Flowchart / Lifecycle

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2. Interrupt Processing Cycle Steps

When an interrupt is detected at the end of the execute cycle:

1. Push

Q3. Explain the working of a microprogrammed (soft-wired) control unit with a block diagram and example (ADD AX, BX).

1. Concept & Block Diagram

A Microprogrammed Control Unit generates control signals by reading consecutive microinstructions from a specialized Control Memory (ROM).

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2. Working Step-by-Step for `ADD AX, BX`

1. Opcode of `ADD` in IR is mapped to Control Memory starting address (e.g., `Address 100`).

2. CAR

Q4. Explain the working of a hardwired control unit with a block diagram and example (ADD AX, BX).

1. Concept & Block Diagram

A Hardwired Control Unit uses fixed logic gates, decoders, and a step counter to generate control signals directly in real-time.

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2. Working for `ADD AX, BX`

1. Opcode Decoder identifies `ADD` instruction.

2. Step

Q5. Compare hardwired and microprogrammed control units in detail.

| Comparison Parameter | Hardwired Control Unit | Microprogrammed Control Unit |

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Q6. Explain microinstruction sequencing and execution with a suitable example (ADD AX, BX).

1. Sequencing & Execution Mechanism

Microinstruction sequencing determines the Control Memory address of the *next* microinstruction (Increment CAR, Branch on Flag, or Map Opcode).

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* **Step 1 (100):** Transfers `AX` contents to temporary register. Next address = 101.

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Q7. Explain Flynn's classification of computer architecture with diagrams and real-life analogies.

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MISD (Multiple Inst, Single Data) MIMD (Multiple Inst, Multiple Data)

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1. SISD: Uniprocessor. Single instruction on single data.

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Q8. Explain instruction pipelining with a 5-stage pipeline timing diagram; calculate total cycles.

1. 5-Stage Pipeline Timing Diagram (4 Instructions: I1 to I4)

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2. Cycle Calculation for N = 4 instructions, K = 5 stages

* **Non-Pipelined Cycles:** $N \times K = 4 \times 5 = 20$ cycles**.

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Q9. Explain the three types of pipeline hazards (structural, data, control) with examples and their solutions.

1. Structural Hazard: Hardware resource conflict (e.g., IF and MEM stages accessing single RAM simultaneously).

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Q10. Explain pipeline hazard handling techniques: stalling, forwarding, branch prediction, and compiler scheduling.

1. Stalling (Bubbles): Inserts NOP (No Operation) cycles into the pipeline, delaying dependent stages until hazard resolves.

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